

These are the notes for Week 2 for Max's and Sarah's content. They cover the infinite volume of Φ_2^4 and the construction of Φ_3^4 .

We assume that in Week 1 the standard Besov spaces, paraproducts and extension operators have been covered and besov estimates. If not it will be covered at the beginning.

Tentative plan

(Lectures and TA Sessions are one hour each)

- Lecture 1. Section 1.1. The infinite volume limit in the DPD decomposition. This requires weighted estimates and weighted Besov spaces. They have already seen the estimates in standard Besov spaces. Discuss the stochastic objects (this could give motivation for the specific Besov setting)
- Lecture 2. Start Φ_3^4 . Paracontrolled expansion.
- Lecture 3. The second renormalization, commutator lemmas.
- Lecture 4. Tightness in three dimensions and infinite volume limit. Some properties of the measure.
- Lecture 5. Stochastic estimates for one of the trees. Discuss the Wick cube and its regularity. Also an L^2 computation with higher order trees. (Add something to the notes) [If time allows sketch the coming-down from infinity method in two dimensions from the paper [3]].
- TA Session 1. Characterisation of Besov functions as sum of spectrally limited functions. Paraproducts.
- TA Session 2. Weighted Besov spaces, estimates in weighted spaces.
- TA Session 3. Wavelet representation and the localization operators.
- TA Session 4. Approximate duality and commutator lemma.

Chapter 1

A PDE construction of the Euclidean Φ_3^4 theory

Based on arguments from the theory of PDEs, we intend to construct the Euclidean Φ^4 quantum field theory on $\mathbb{R}^3$. This is a probability measure ν on the space of Schwartz distributions $\mathcal{S}'(\mathbb{R}^3)$ which is formally represented by

$$\nu(d\varphi) \sim \exp \left\{ -2 \int_{\mathbb{R}^3} \left[\frac{\lambda}{4} |\varphi(x)|^4 + \frac{m^2}{2} |\varphi(x)|^2 + \frac{1}{2} |\nabla \varphi(x)|^2 \right] dx \right\} d\varphi,$$

where $m^2 \in \mathbb{R}$ is the mass and $\lambda > 0$ is the coupling constant. The above expression is only formal, because powers of distributions are not well defined in general. The measure ν is associated to the corresponding stochastic quantization equation

$$(\partial_t + m^2 - \Delta) \varphi + \lambda \varphi^3 - \infty \varphi = \xi, \quad (t, x) \in \mathbb{R}_+ \times \mathbb{R}^3, \quad (1.1)$$

where ξ is a space-time white noise on $\mathbb{R}^3$ and the term $-\infty \varphi$ stands for the so-called mass renormalization. This permits to make sense of the nonlinearity in (1.1) and in turn to obtain an honest probability measure ν as the invariant measure of (1.1).

For notational simplicity, we let $m^2 = \lambda = 1$ in the sequel. For $d \in \{2, 3\}$, let $\Lambda_\varepsilon = (\varepsilon \mathbb{Z})^d$ for $\varepsilon = 2^{-N}$, $N \in \mathbb{N}_0$, be the rescaled lattice $\mathbb{Z}^d$ and let $\Lambda_{M,\varepsilon} = ((\varepsilon \mathbb{Z}) / (M \mathbb{Z}))^d$ be the periodic lattice with mesh size ε and side length M with $M / (2\varepsilon) \in \mathbb{N}$. We denote by Δ_ε the lattice Laplacian on Λ_ε

$$\Delta_\varepsilon f(x) := \varepsilon^{-2} \sum_{i=1}^d (f(x + \varepsilon e_i) - 2f(x) + f(x - \varepsilon e_i)), \quad x \in \Lambda_\varepsilon,$$

where $(e_i)_{i=1,\dots,d}$ is the canonical basis of $\mathbb{R}^d$. Let $Q_\varepsilon := 1 - \Delta_\varepsilon$, $\mathcal{L}_\varepsilon = \partial_t + Q_\varepsilon$ and we write L for the continuum analogue of $\mathcal{L}_\varepsilon$.

We approximate the stochastic quantization equation (1.1) by the finite dimensional lattice model

$$\mathcal{L}_\varepsilon \varphi_{M,\varepsilon} + \varphi_{M,\varepsilon}^3 - 3c_{M,\varepsilon} \varphi_{M,\varepsilon} = \xi_{M,\varepsilon}, \quad x \in \Lambda_{M,\varepsilon}. \quad (1.2)$$

Here $c_{M,\varepsilon}$ are renormalization constants diverging as $M \rightarrow \infty$ and $\varepsilon \rightarrow 0$ and $\xi_{M,\varepsilon}$ is the discrete and periodic approximation of a space-time white noise ξ on $\mathbb{R}^d$

$$\xi_{M,\varepsilon}(t, x) := \varepsilon^{-d} \langle \xi_M(t, \cdot), \mathbf{1}_{|\cdot - x| \leq \varepsilon/2} \rangle, \quad (t, x) \in \mathbb{R} \times \Lambda_{M,\varepsilon},$$

with ξ_M being the periodization of ξ

$$\xi_M(h) := \xi(h_M), \quad \text{where } h_m(t, x) := \mathbf{1}_{\left[-\frac{M}{2}, \frac{M}{2}\right]^d}(x) \sum_{y \in M\mathbb{Z}^d} h(t, x + y), \quad h \in L^2(\mathbb{R} \times \mathbb{R}^d).$$

Then (1.2) is a finite-dimensional SDE in a gradient form and it has a unique invariant measure $\nu_{M,\varepsilon}$ given by

$$\nu_{M,\varepsilon}(d\varphi) \sim \exp \left\{ -2\varepsilon^d \sum_{x \in \Lambda_{M,\varepsilon}} \left[\frac{1}{4}|\varphi_x|^4 + \frac{-3c_{M,\varepsilon} + 1}{2}|\varphi_x|^2 + \frac{1}{2}|\nabla_\varepsilon \varphi_x|^2 \right] \right\} \prod_{x \in \Lambda_{M,\varepsilon}} d\varphi_x, \quad (1.3)$$

where ∇_ε denotes the discrete gradient

$$\nabla_\varepsilon f(x) := \left(\frac{f(x + \varepsilon e_i) - f(x)}{\varepsilon} \right)_{i=1,\dots,d}$$

satisfying the corresponding integration by parts

$$\langle \Delta_\varepsilon f, g \rangle_{M,\varepsilon} = -\langle \nabla_\varepsilon f, \nabla_\varepsilon g \rangle_{M,\varepsilon},$$

where by $\langle \cdot, \cdot \rangle_{M,\varepsilon}$ we denoted the duality associated to $L^2(\Lambda_{M,\varepsilon})$.

Global existence of solutions to (1.2) can be proved along the lines of Khasminskii nonexplosion test [5, Theorem 3.5] whereas invariance of the measure (1.3) follows from Itô's formula together with the integration by parts.

Our goal is to show that there exists a choice of the renormalization constants $(c_{M,\varepsilon})_{M,\varepsilon}$ such that the family of probability measures $(\nu_{M,\varepsilon})_{M,\varepsilon}$ extended to $\mathcal{S}'(\mathbb{R}^d)$ is tight. Consequently, there exists a subsequence converging weakly in the sense of probability measures to some ν and this is the candidate for the Euclidean quantum field theory on $\mathbb{R}^d$. It can be additionally proved that every such accumulation point ν is translation invariant, reflection positive and non-Gaussian and satisfies a stretched exponential integrability. In other words, it satisfies part of the Osterwalder–Schader axioms for a Euclidean quantum field theory. Furthermore, an integration by parts formula can be established and it leads to the hierarchy of Dyson–Schwinger equations for Euclidean correlation functions.

1.1 Construction of the Euclidean Φ_2^4 theory

Exercise 1.1. Study Section A.2.

In order to explain the main ideas of the construction in a simpler setting, we restrict ourselves now to $d = 2$. Our aim is to find a decomposition of (1.2) where all the quantities can be controlled uniformly in M, ε . In the first step, we isolate the term coming from the noise as it is expected to be the most irregular in the limit $M \rightarrow \infty, \varepsilon \rightarrow 0$. Recall that in $d = 2$, $(\xi_{M,\varepsilon})_{M,\varepsilon}$ only has uniform bounds in a weighted Besov space of regularity $-2 - \kappa$ for every $\kappa > 0$. This regularity becomes worse in 3 dimensions and we will see later that creates further difficulties and additional ideas are needed to complete the proof. To remove this irregularity, we solve the corresponding linear counterpart of (1.2) and let $X_{M,\varepsilon}$ be its stationary solution, that is,

$$\mathcal{L}_\varepsilon X_{M,\varepsilon} = \xi_{M,\varepsilon}. \quad (1.4)$$

By Schauder estimates, $(X_{M,\varepsilon})_{M,\varepsilon}$ is bounded uniformly in the weighted function space $CC^{-\kappa}(\rho^\sigma)$ a.s. for $\kappa, \sigma > 0$ arbitrary and ρ as in (A.2).

Decomposing $\varphi_{M,\varepsilon} = X_{M,\varepsilon} + \eta_{M,\varepsilon}$ we expect $\eta_{M,\varepsilon}$ to be more regular. It satisfies

$$\mathcal{L}_\varepsilon \eta_{M,\varepsilon} + (X_{M,\varepsilon}^3 - 3c_{M,\varepsilon} X_{M,\varepsilon}) + 3(X_{M,\varepsilon}^2 - c_{M,\varepsilon}) \eta_{M,\varepsilon} + 3X_{M,\varepsilon} \eta_{M,\varepsilon}^2 + \eta_{M,\varepsilon}^3 = 0. \quad (1.5)$$

The advantage is that choosing $c_{M,\varepsilon} := \mathbb{E}[X_{M,\varepsilon}^2(t)]$ the terms

$$\llbracket X_{M,\varepsilon}^2 \rrbracket := X_{M,\varepsilon}^2 - c_{M,\varepsilon}, \quad \llbracket X_{M,\varepsilon}^3 \rrbracket := X_{M,\varepsilon}^3 - 3c_{M,\varepsilon}X_{M,\varepsilon}, \quad (1.6)$$

are bounded uniformly in M, ε in $CC^{-\kappa}(\rho^\sigma)$ a.s. In particular, the terms defined in (1.6) are the second and third Wick power of the Gaussian random variable $X_{M,\varepsilon}$. Then by Schauder estimates we expect $(\eta_{M,\varepsilon})_{M,\varepsilon}$ to be bounded uniformly in $CC^{2-\kappa}(\rho^\sigma)$, hence it is function valued and all the products in (1.5) are well-defined.

1.1.1 Weighted energy estimate

As the next step, we want to derive a weighted energy estimate for $\eta_{M,\varepsilon}$ and make it uniform in M, ε . To this end, we test (1.5) by $\rho^4 \eta_{M,\varepsilon}$, or alternatively we apply the chain rule to calculate $\frac{1}{2} \partial_t \|\rho^2 \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2$. We obtain

$$\begin{aligned} & \frac{1}{2} \partial_t \|\rho^2 \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho^2 \nabla_\varepsilon \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 \\ &= -\langle \rho^4 \eta_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^3 \rrbracket \rangle_\varepsilon + 3 \llbracket X_{M,\varepsilon}^2 \rrbracket \eta_{M,\varepsilon} + 3X_{M,\varepsilon} \eta_{M,\varepsilon}^2 \rangle_\varepsilon. \end{aligned}$$

The $L^{4,\varepsilon}$ -norm on the left hand side is obtained from the cubic term, the $L^{2,\varepsilon}$ -norm from the massive term $-\eta_{M,\varepsilon}$, and finally the norm of the gradient comes after integration by parts from the term $-\Delta_\varepsilon \eta_{M,\varepsilon}$. Observe that by Lemma A.1 and Lemma A.5, the two last terms on the left hand side can be estimated from below by the $H^{1-\kappa}(\rho^2)$ -norm of $\eta_{M,\varepsilon}$ for $\kappa > 0$ small so that the above rewrites as

$$\frac{1}{2} \partial_t \|\rho^2 \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2 \quad (1.7)$$

$$\lesssim -\langle \rho^4 \eta_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^3 \rrbracket \rangle_\varepsilon + 3 \llbracket X_{M,\varepsilon}^2 \rrbracket \eta_{M,\varepsilon} + 3X_{M,\varepsilon} \eta_{M,\varepsilon}^2 \rangle_\varepsilon. \quad (1.8)$$

All the terms on the left hand side will be crucially used to control the right hand side. More precisely, it is necessary to absorb into the left hand side the norms of $\eta_{M,\varepsilon}$ needed in the estimates of the right hand side. The final estimate can only depend in a polynomial way on the given data, namely, the uniform in M, ε estimates of

$$\mathbb{X}_{M,\varepsilon} := (X_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^2 \rrbracket, \llbracket X_{M,\varepsilon}^3 \rrbracket) \in [C_T C^{-\kappa,\varepsilon}(\rho^\sigma)]^3$$

for $\kappa, \sigma > 0$ small and an arbitrary $T \in (0, \infty)$. This is achieved by combining the basic results for discrete Besov space from Section A.1 with weighted Young inequality in a suitable way. The key role is especially played by the L^4 -norm because it permits to balance the loss of weight on the right hand side. This is due to the fact that the noise terms $\mathbb{X}_{M,\varepsilon}$ always require a part of the weight to be bounded uniformly in M, ε .

For the sequel, we fix the parameters $\kappa, \sigma > 0$ small as well as a weight ρ as in (A.2) and a small parameter $\iota \in (0, 1)$ so that $\rho^\iota \in L^{4,0}$. This technical assumption will help us with several embeddings to close the desired energy estimate.

First, we apply Lemma A.1, the duality (Lemma A.2), the embedding $H^{1-\kappa,\varepsilon}(\rho^2) = B_{2,2}^{1-\kappa} \subset B_{1,1}^\kappa(\rho^{4-\sigma})$ which holds provided $\rho^{2-\sigma} \in L^{2,\varepsilon}$ and the weighted Young inequality, to bound for an arbitrary $\delta \in (0, 1)$

$$\begin{aligned} & |\langle \rho^4 \eta_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^3 \rrbracket \rangle_\varepsilon| \lesssim \|\rho^\sigma \llbracket X_{M,\varepsilon}^3 \rrbracket\|_{C^{-\kappa,\varepsilon}} \|\rho^{4-\sigma} \eta_{M,\varepsilon}\|_{B_{1,1}^{\kappa,\varepsilon}} \\ & \lesssim \|\rho^\sigma \llbracket X_{M,\varepsilon}^3 \rrbracket\|_{C^{-\kappa,\varepsilon}} \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}} \leq Q(\mathbb{X}_{M,\varepsilon}) + \delta \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2. \end{aligned} \quad (1.9)$$

Here and in the sequel, $Q(\mathbb{X}_{M,\varepsilon})$ always denotes a polynomial in the uniform norms of $\mathbb{X}_{M,\varepsilon}$ and it may change from line to line. Due to the small constant δ , the last term on the right hand side of (1.9) can be absorbed into (1.7).

Starting similarly for the second term in (1.8) and using the estimate for powers from Lemma A.7 as well as the interpolation from Lemma A.3 with $\theta = \frac{1-3\kappa}{1-\kappa}$, the embedding from Lemma A.6, we obtain

$$\begin{aligned}
|\langle \rho^4 \eta_{M,\varepsilon}, 3[X_{M,\varepsilon}^2] \eta_{M,\varepsilon} \rangle_\varepsilon| &\lesssim \|\rho^\sigma [X_{M,\varepsilon}^2]\|_{C^{-\kappa,\varepsilon}} \|\rho^{4-\sigma} \eta_{M,\varepsilon}^2\|_{B_{1,1}^{\kappa,\varepsilon}} \\
&\lesssim \|\rho^\sigma [X_{M,\varepsilon}^2]\|_{C^{-\kappa,\varepsilon}} \|\rho^{1+\iota} \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}} \|\rho^{3-\iota-\sigma} \eta_{M,\varepsilon}\|_{H^{2\kappa,\varepsilon}} \\
&\lesssim \|\rho^\sigma [X_{M,\varepsilon}^3]\|_{C^{-\kappa,\varepsilon}} \|\rho^{1+\iota} \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^{1+\theta} \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^{1-\theta} \\
&\lesssim \|\rho^\sigma [X_{M,\varepsilon}^3]\|_{C^{-\kappa,\varepsilon}} \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^{1+\theta} \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^{1-\theta} \\
&\leq Q(\mathbb{X}_{M,\varepsilon}) + \delta(\|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2).
\end{aligned}$$

In the same spirit, the last term is bounded as

$$\begin{aligned}
|\langle \rho^4 \eta_{M,\varepsilon}, 3X_{M,\varepsilon} \eta_{M,\varepsilon}^2 \rangle_\varepsilon| &\lesssim \|\rho^\sigma X_{M,\varepsilon}\|_{C^{-\kappa,\varepsilon}} \|\rho^{4-\sigma} \eta_{M,\varepsilon}^3\|_{B_{1,1}^{\kappa,\varepsilon}} \\
&\lesssim \|\rho^\sigma X_{M,\varepsilon}\|_{C^{-\kappa,\varepsilon}} \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^2 \|\rho^{2-\sigma} \eta_{M,\varepsilon}\|_{H^{2\kappa,\varepsilon}} \\
&\lesssim \|\rho^\sigma X_{M,\varepsilon}\|_{C^{-\kappa,\varepsilon}} \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^2 \|\rho^{1+\iota} \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^\theta \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^{1-\theta} \\
&\leq Q(\mathbb{X}_{M,\varepsilon}) + \delta(\|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2).
\end{aligned}$$

Finally, this brings us to the estimate

$$\frac{1}{2} \partial_t \|\rho^2 \eta_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2 \leq Q(\mathbb{X}_{M,\varepsilon}) \quad (1.10)$$

which we exploit further in the next section in order to deduce tightness of the approximate invariant measures (1.3).

1.1.2 Tightness

Exercise 1.2. Study Section A.2.

Let $\varphi_{M,\varepsilon}$ and $X_{M,\varepsilon}$ be stationary solutions to (1.2) and (1.4), respectively, defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. These solutions can be obtained by solving (1.2) and (1.4) with random initial conditions sampled from the probability measures (1.3) and

$$\frac{1}{Z_{M,\varepsilon}} \exp \left\{ -2\varepsilon^d \sum_{x \in \Lambda_{M,\varepsilon}} \left[\frac{-3c_{M,\varepsilon} + 1}{2} |\varphi_x|^2 + \frac{1}{2} |\nabla_\varepsilon \varphi_x|^2 \right] \right\} \prod_{x \in \Lambda_{M,\varepsilon}} d\varphi_x,$$

respectively, where $Z_{M,\varepsilon}$ denotes a normalizing constant. As above, we define $\eta_{M,\varepsilon} := \varphi_{M,\varepsilon} - X_{M,\varepsilon}$ and observe that it is a stationary solution to (1.5) so that the weighted energy estimate (1.10) holds true. Taking expectation and using stationarity therefore implies

$$\mathbb{E} \|\rho \eta_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \mathbb{E} \|\rho^2 \eta_{M,\varepsilon}\|_{H^{1-\kappa,\varepsilon}}^2 \leq \mathbb{E} Q(\mathbb{X}_{M,\varepsilon}) \lesssim 1, \quad (1.11)$$

where the implicit constant is independent of M, ε . In particular, it is important to note that the form of the polynomial Q does not depend on M, ε : the only dependence of $Q(\mathbb{X}_{M,\varepsilon})$ on these parameters comes through the norms of $\mathbb{X}_{M,\varepsilon}$ and the quantity is uniformly bounded in expectation.

All the involved objects $\varphi_{M,\varepsilon}$, $X_{M,\varepsilon}$, $\eta_{M,\varepsilon}$ are extended periodically to the full lattice Λ_ε . In order to establish tightness of the measures (1.3), that is, the laws of $\varphi_{M,\varepsilon}$ at an arbitrary time, we make use of the extension operator $\mathcal{E}^\varepsilon$ defined in Appendix A.2, which further permits to extend $\varphi_{M,\varepsilon}$ as a distribution on Λ_ε to a distribution on the full space $\mathbb{R}^2$.

Theorem 1.1. *The family of laws of $(\mathcal{E}^\varepsilon \varphi_{M,\varepsilon}(0))_{M,\varepsilon}$ is tight on $C^{-2\kappa}(\rho^{2+\kappa})$. Every accumulation point ν is the candidate Euclidean Φ_2^4 quantum field theory.*

Proof. Lemma A.9 and (1.11) lead to

$$\mathbb{E}\|\mathcal{E}^\varepsilon X_{M,\varepsilon}(0)\|_{C^{-\kappa}(\rho^\sigma)}^2 \lesssim 1, \quad \mathbb{E}\|\mathcal{E}^\varepsilon \eta_{M,\varepsilon}(0)\|_{H^{1-\kappa,\varepsilon}(\rho^2)}^2 \lesssim 1$$

and consequently

$$\mathbb{E}\|\mathcal{E}^\varepsilon \varphi_{M,\varepsilon}(0)\|_{C^{-\kappa}(\rho^2)}^2 \lesssim \mathbb{E}\|\mathcal{E}^\varepsilon X_{M,\varepsilon}(0)\|_{C^{-\kappa}(\rho^\sigma)}^2 + \mathbb{E}\|\mathcal{E}^\varepsilon \eta_{M,\varepsilon}(0)\|_{H^{1-\kappa,\varepsilon}(\rho^2)}^2 \lesssim 1.$$

Since the embedding $C^{-\kappa}(\rho^2) \subset C^{-2\kappa}(\rho^{2+\kappa})$ is compact, the result follows. $\square$

1.2 Construction of the Euclidean Φ_3^4 theory

Exercise 1.3. Study Section A.3, Section A.4 and Section A.5.

1.2.1 Decomposition

Compared to the 2 dimensional setting, the situation is significantly more involved in 3 spatial dimension. The space-time white noise ξ is a random distribution of regularity $-5/2 - \kappa$, $\kappa > 0$, and therefore the solution $X_{M,\varepsilon}$ to the linear counterpart (1.4) only has uniform bounds in $CC^{-1/2-\kappa}(\rho^\sigma)$, $\kappa, \sigma > 0$. Its second and third Wick powers $\llbracket X_{M,\varepsilon}^2 \rrbracket$ and $\llbracket X_{M,\varepsilon}^3 \rrbracket$ are worse, namely, they have the regularities $-1 - \kappa$ and $-3/2 - \kappa$, respectively. Moreover, the limit $\llbracket X^3 \rrbracket$ only makes sense as space-time distribution and cannot therefore be evaluated at a fixed time t . With this power counting at hand, the uniform regularity of $\eta_{M,\varepsilon}$ solving (1.5) expected from the Schauder estimates is $CC^{1/2-\kappa}$. Consequently, (1.5) contains products $\llbracket X_{M,\varepsilon}^2 \rrbracket \eta_{M,\varepsilon}$ and $X_{M,\varepsilon} \eta_{M,\varepsilon}$ that are ill-defined in the limit $M \rightarrow \infty$, $\varepsilon \rightarrow 0$.

As a consequence, a more involved renormalization is necessary and the renormalization constants in the dynamics (1.2) is of the form $c_{M,\varepsilon} = a_{M,\varepsilon} - b_{M,\varepsilon}$ with $a_{M,\varepsilon} = \mathbb{E}[X_{M,\varepsilon}^2(t)]$ and $b_{M,\varepsilon}$ to be determined below. The usual way how to proceed is to continue with the decomposition and again isolate the most irregular term, letting $X_{M,\varepsilon}^\Psi$ be a stationary solution to

$$\mathcal{L}_\varepsilon X_{M,\varepsilon}^\Psi = \llbracket X_{M,\varepsilon}^3 \rrbracket.$$

This leads to the decomposition $\varphi_{M,\varepsilon} = X_{M,\varepsilon} - X_{M,\varepsilon}^\Psi + \zeta_{M,\varepsilon}$ and cancels $\llbracket X_{M,\varepsilon}^3 \rrbracket$ from the relevant dynamics. More precisely, the most irregular term in the dynamics of the remainder $\zeta_{M,\varepsilon}$ is the paraproduct $\llbracket X_{M,\varepsilon}^2 \rrbracket \triangleright X_{M,\varepsilon}^\Psi$ which can be bounded uniformly in $CC^{-1-\kappa}(\rho^\sigma)$. This is however not sufficient for the intended weighted energy estimate in the spirit of Section 1.1.1, because the expected uniform regularity of $\zeta_{M,\varepsilon}$ is only two degrees better, i.e. $CC^{1-\kappa}(\rho^\sigma)$, leading to the slightly problematic inner product $\langle \rho^4 \zeta_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^2 \rrbracket \triangleright X_{M,\varepsilon}^\Psi \rangle_\varepsilon$.

To solve this problem, we proceed differently and instead of $X_{M,\varepsilon}^\Psi$ we remove $Y_{M,\varepsilon}$ which is a solution to

$$\mathcal{L}_\varepsilon Y_{M,\varepsilon} = -[\![X_{M,\varepsilon}^3]\!] - 3(U_\varepsilon^\varepsilon[\![X_{M,\varepsilon}^2]\!]) > Y_{M,\varepsilon}, \quad Y_{M,\varepsilon}(0) = -X_{M,\varepsilon}^\Psi(0),$$

where $U_\varepsilon^\varepsilon$ is the localization operator defined in Section A.4. In other words, $Y_{M,\varepsilon}$ also contains the most irregular part of the product $[\![X_{M,\varepsilon}^2]\!] > X_{M,\varepsilon}^\Psi$. With a suitable choice of the constant $L = L(M, \varepsilon)$ determining $U_\varepsilon^\varepsilon$ (cf. Lemma A.13) we are able to construct the unique solution to this problem via Banach's fixed point theorem. Consequently, we find our decomposition $\varphi_{M,\varepsilon} = X_{M,\varepsilon} + Y_{M,\varepsilon} + \phi_{M,\varepsilon}$ together with the dynamics for the remainder

$$\mathcal{L}_\varepsilon \phi_{M,\varepsilon} + \phi_{M,\varepsilon}^3 = -3[\![X_{M,\varepsilon}^2]\!] > \phi_{M,\varepsilon} - 3[\![X_{M,\varepsilon}^2]\!] \circ \phi_{M,\varepsilon} - 3b_{M,\varepsilon} \phi_{M,\varepsilon} + \Xi_{M,\varepsilon}. \quad (1.12)$$

The first term on the right hand side is the most irregular contribution, the second term is not controlled uniformly in M, ε , the third term is needed for the renormalization and $\Xi_{M,\varepsilon}$ contains various terms that are more regular and in principle not problematic or that can be constructed as stochastic objects using the remaining counterterm $-3b_{M,\varepsilon}(X_{M,\varepsilon} + Y_{M,\varepsilon})$.

The advantage of this decomposition with $\phi_{M,\varepsilon}$ as opposed to the usual approach leading to $\zeta_{M,\varepsilon}$ above is that together with $[\![X_{M,\varepsilon}^3]\!]$ we cancelled also the second most irregular contribution $(U_\varepsilon^\varepsilon[\![X_{M,\varepsilon}^2]\!]) > Y_{M,\varepsilon}$, which is too irregular to be controlled as a forcing f using the energy method. The same difficulty of course comes with $[\![X_{M,\varepsilon}^2]\!] > \phi_{M,\varepsilon}$ in (1.12), however, since it depends on the solution $\phi_{M,\varepsilon}$ we are able to control it using a paracontrolled ansatz. To explain this, let us also turn our attention to the resonant product $[\![X_{M,\varepsilon}^2]\!] \circ \phi_{M,\varepsilon}$ which poses problems as well. When applying the energy method to (1.12), these two terms appear in the form

$$\langle \rho^4 \phi_{M,\varepsilon}, -3[\![X_{M,\varepsilon}^2]\!] \circ \phi_{M,\varepsilon} \rangle_\varepsilon + \langle \rho^4 \phi_{M,\varepsilon}, -3[\![X_{M,\varepsilon}^2]\!] > \phi_{M,\varepsilon} \rangle_\varepsilon,$$

where we included a polynomial weight ρ as in (A.2). The key observation is that the presence of the duality product permits to show that these two terms *approximately* coincide, in the sense that their difference denoted by $D_{\rho^4,\varepsilon}(\phi_{M,\varepsilon}, -3[\![X_{M,\varepsilon}^2]\!], \phi_{M,\varepsilon})$ is controlled by the expected uniform bounds. This is proven generally in Lemma A.14. As a consequence, we obtain

$$\begin{aligned} & \frac{1}{2} \partial_t \|\rho^2 \phi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \phi_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \langle \rho^4 \phi_{M,\varepsilon}, Q_\varepsilon \phi_{M,\varepsilon} \rangle_\varepsilon \\ &= \langle \rho^4 \phi_{M,\varepsilon}, -3 \cdot 2[\![X_{M,\varepsilon}^2]\!] > \phi_{M,\varepsilon} \rangle_\varepsilon + D_{\rho^4,\varepsilon}(\phi_{M,\varepsilon}, -3[\![X_{M,\varepsilon}^2]\!], \phi_{M,\varepsilon}) + \Xi_{M,\varepsilon}. \end{aligned}$$

Finally, since the last term on the left hand side as well as the first term on the right hand side are diverging, the idea is to couple them by the following paracontrolled ansatz. We define

$$Q_\varepsilon \psi_{M,\varepsilon} := Q_\varepsilon \phi_{M,\varepsilon} + 3[\![X_{M,\varepsilon}^2]\!] > \phi_{M,\varepsilon}$$

and expect that the sum of the two terms on the right hand side is more regular than each of them separately. In other words, $\psi_{M,\varepsilon}$ is uniformly more regular than $\phi_{M,\varepsilon}$. Indeed, with this ansatz we may complete the square and obtain a formula analogous to (1.7), (1.8) where all the terms on the right hand side can be controlled.

However, the formula in 3 dimensions is much more complicated. Before we can spell it out, we need to introduce further stochastic objects, which can be constructed similarly to the usual Φ_3^4 terms, see e.g. [3]. Namely, we define

$$\begin{aligned} X_{M,\varepsilon}^\Psi &:= L_\varepsilon^{-1}[\![X_{M,\varepsilon}^2]\!], & X_{M,\varepsilon}^\Psi &:= X_{M,\varepsilon} \circ X_{M,\varepsilon}^\Psi, \\ X_{M,\varepsilon}^\Psi &:= 9[\![X_{M,\varepsilon}^2]\!] \circ Q_\varepsilon^{-1}[\![X_{M,\varepsilon}^2]\!] - 3b_{M,\varepsilon}, \\ \tilde{X}_{M,\varepsilon}^\Psi &:= 9[\![X_{M,\varepsilon}^2]\!] \circ X_{M,\varepsilon}^\Psi - 3\tilde{b}_{M,\varepsilon}(t), & X_{M,\varepsilon}^\Psi &:= 3[\![X_{M,\varepsilon}^2]\!] \circ X_{M,\varepsilon}^\Psi - 3b_{M,\varepsilon} X_{M,\varepsilon}, \end{aligned}$$

where $b_{M,\varepsilon}, \tilde{b}_{M,\varepsilon}(t)$ are suitable renormalization constants. It follows from standard estimates that $|\tilde{b}_{M,\varepsilon}(t) - b_{M,\varepsilon}| \lesssim |\log t|$ uniformly in M, ε . Altogether, the relevant stochastic objects are

$$\mathbb{X}_{M,\varepsilon} := (X_{M,\varepsilon}, \llbracket X_{M,\varepsilon}^2 \rrbracket, X_{M,\varepsilon}^\Psi, X_{M,\varepsilon}^{\check{\Psi}}, X_{M,\varepsilon}^{\check{\Psi}}, \tilde{X}_{M,\varepsilon}^{\check{\Psi}}, X_{M,\varepsilon}^{\check{\Psi}})$$

and the norms in the following function spaces are controlled uniformly in M, ε and will be used in our energy estimate

$$\begin{aligned} X_{M,\varepsilon} &\in C_T C^{-1/2-\kappa,\varepsilon}(\rho^\sigma), & \llbracket X_{M,\varepsilon}^2 \rrbracket &\in C_T C^{-1-\kappa,\varepsilon}(\rho^\sigma), \\ X_{M,\varepsilon}^\Psi &\in C_T C^{1/2-\kappa,\varepsilon}(\rho^\sigma) \cap C_T^{\beta/2} L^{\infty,\varepsilon}(\rho^\sigma), \\ X_{M,\varepsilon}^{\check{\Psi}} &\in C_T C^{-\kappa,\varepsilon}(\rho^\sigma), & X_{M,\varepsilon}^{\check{\Psi}} &\in C_T C^{-\kappa,\varepsilon}(\rho^\sigma), \\ \tilde{X}_{M,\varepsilon}^{\check{\Psi}} &\in C_T C^{-\kappa,\varepsilon}(\rho^\sigma), & X_{M,\varepsilon}^{\check{\Psi}} &\in C_T C^{-1/2-\kappa,\varepsilon}(\rho^\sigma), \end{aligned}$$

where $\kappa, \sigma > 0, \beta > 3\kappa$ and the time horizon $T \in (0, \infty)$ is fixed but arbitrary at this point. It will be determined in Theorem 1.3.

As in Section 1.1, $Q(\mathbb{X}_{M,\varepsilon}) = Q_\rho(\mathbb{X}_{M,\varepsilon})$ denotes a generic polynomial in the above norms, whose coefficients depend on ρ but are independent of M, ε and change from line to line.

With all these preparations at hand, we are finally able to write down the 3 dimensional counterpart of (1.7), (1.8). It reads as

$$\frac{1}{2} \partial_t \|\rho^2 \phi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \phi_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \psi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho^2 \nabla_\varepsilon \psi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 = \Theta_{M,\varepsilon} + \Psi_{M,\varepsilon}, \quad (1.13)$$

where

$$\begin{aligned} \Theta_{M,\varepsilon} &:= -\langle [\nabla_\varepsilon, \rho^4] \psi_{M,\varepsilon}, \nabla_\varepsilon \psi_{M,\varepsilon} \rangle_\varepsilon + \langle [Q_\varepsilon, \rho^4] Q_\varepsilon^{-1} [3 \llbracket X_{M,\varepsilon}^2 \rrbracket > \phi_{M,\varepsilon}], \psi_{M,\varepsilon} \rangle_\varepsilon \\ &+ \langle \rho^4 \phi_{M,\varepsilon}^2, X_{M,\varepsilon}^{\check{\Psi}} \rangle_\varepsilon + D_{\rho^4,\varepsilon}(\phi_{M,\varepsilon}, -3 \llbracket X_{M,\varepsilon}^2 \rrbracket, \phi_{M,\varepsilon}) + \langle \rho^4 \phi_{M,\varepsilon}, \tilde{C}_\varepsilon(\phi_{M,\varepsilon}, 3 \llbracket X_{M,\varepsilon}^2 \rrbracket, 3 \llbracket X_{M,\varepsilon}^2 \rrbracket) \rangle_\varepsilon \\ &+ D_{\rho^4,\varepsilon}(\phi_{M,\varepsilon}, 3 \llbracket X_{M,\varepsilon}^2 \rrbracket, Q_\varepsilon^{-1} [3 \llbracket X_{M,\varepsilon}^2 \rrbracket > \phi_{M,\varepsilon}]), \\ \Psi_{M,\varepsilon} &:= \langle \rho^4 \phi_{M,\varepsilon}, -3 \llbracket X_{M,\varepsilon}^2 \rrbracket \rangle_\varepsilon + \langle Y_{M,\varepsilon} + \phi_{M,\varepsilon}, -3 X_{M,\varepsilon} (Y_{M,\varepsilon} + \phi_{M,\varepsilon})^2 - Y_{M,\varepsilon}^3 \rangle_\varepsilon \\ &+ \langle \rho^4 \phi_{M,\varepsilon}, -3 Y_{M,\varepsilon}^2 \phi_{M,\varepsilon} - 3 Y_{M,\varepsilon} \phi_{M,\varepsilon}^2 - 3 (U_\varepsilon^\varepsilon \llbracket X_{M,\varepsilon}^2 \rrbracket) > Y_\varepsilon + Z_{M,\varepsilon} \rangle_\varepsilon \end{aligned}$$

and

$$\begin{aligned} Z_{M,\varepsilon} &:= X_{M,\varepsilon}^{\check{\Psi}} + \tilde{X}_{M,\varepsilon}^{\check{\Psi}} Y_{M,\varepsilon} + 3(\tilde{b}_{M,\varepsilon} - b_{M,\varepsilon}) Y_{M,\varepsilon} + \tilde{C}_\varepsilon(Y_{M,\varepsilon}, 3 \llbracket X_{M,\varepsilon}^2 \rrbracket, 3 \llbracket X_{M,\varepsilon}^2 \rrbracket) \\ &- 3 \llbracket X_{M,\varepsilon}^2 \rrbracket \circ L_\varepsilon^{-1} (3 U_\varepsilon^\varepsilon \llbracket X_{M,\varepsilon}^2 \rrbracket > Y_{M,\varepsilon}). \end{aligned}$$

1.2.2 Weighted energy estimate

The goal now is to estimate all the terms on the right hand side of (1.13). The key ideas already appeared in the 2 dimensional setting presented in Section 1.1.1. To summarize:

- $\phi_{M,\varepsilon}$ and $\psi_{M,\varepsilon}$ can only be estimated in the norms appearing on the left hand side of (1.13),
- the stochastic objects can only be estimated in the norms that are uniform in M, ε in expectation as presented above,
- we shall use interpolation and weighted Young inequality to absorb the norms of $\phi_{M,\varepsilon}$ and $\psi_{M,\varepsilon}$ into the left hand side,
- we shall use the continuity of the involved commutators presented in Section A.5.

Working carefully term by term, we eventually conclude the desired result.

Theorem 1.2. *Let ρ be a weight such that $\rho^\iota \in L^{4,0}$ for some $\iota \in (0, 1)$. Then for some $\theta \in (0, 1)$ it holds*

$$\begin{aligned} & \frac{1}{2} \partial_t \|\rho^2 \phi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho \phi_{M,\varepsilon}\|_{L^{4,\varepsilon}}^4 + \|\rho^2 \psi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho^2 \nabla_\varepsilon \psi_{M,\varepsilon}\|_{L^{2,\varepsilon}}^2 + \|\rho^2 \phi_{M,\varepsilon}\|_{H^{1-2\kappa,\varepsilon}}^2 \\ & \leq (1 + |\log t|^{4/(2+\theta)}) Q(\mathbb{X}_{M,\varepsilon}). \end{aligned}$$

1.2.3 Tightness

Compared to the 2 dimensional setting, in 3 dimensions there are a number of subtleties appearing also in the tightness argument. Recall that we decomposed $\varphi_{M,\varepsilon} = X_{M,\varepsilon} + Y_{M,\varepsilon} + \phi_{M,\varepsilon}$ and we introduced an additional paracontrolled ansatz of the form

$$\psi_{M,\varepsilon} = \phi_{M,\varepsilon} + Q_\varepsilon^{-1}[3\llbracket X_{M,\varepsilon}^2 \rrbracket > \phi_{M,\varepsilon}].$$

Let now $(\varphi_{M,\varepsilon}, X_{M,\varepsilon})$ be jointly stationary. Nevertheless, $Y_{M,\varepsilon}$ solves an initial value problem and is therefore not stationary. Accordingly, also $\phi_{M,\varepsilon}$ and $\psi_{M,\varepsilon}$ are not stationary, but they are the quantities for which the energy estimate from Theorem 1.2 holds. On the other hand, we may consider $X_{M,\varepsilon}^\Psi$ stationary so that the triple $(\varphi_{M,\varepsilon}, X_{M,\varepsilon}, X_{M,\varepsilon}^\Psi)$ is jointly stationary and this leads us to another decomposition of $\varphi_{M,\varepsilon}$, namely $\varphi_{M,\varepsilon} = X_{M,\varepsilon} - X_{M,\varepsilon}^\Psi + \zeta_{M,\varepsilon}$ where all the summands are stationary. Recalling the definition of $Y_{M,\varepsilon}$ it holds

$$\zeta_{M,\varepsilon} = -L_\varepsilon^{-1}[3(U_\varepsilon^\Psi \llbracket X^2 \rrbracket) > Y_{M,\varepsilon}] + \phi_{M,\varepsilon}. \quad (1.14)$$

Both expressions for $\varphi_{M,\varepsilon}$ will be used in order to establish the desired tightness of the approximate measures $\nu_{M,\varepsilon}$. More precisely, we prove the tightness of the joint laws of the family of extended distributions $(\mathcal{E}^\varepsilon \varphi_{M,\varepsilon}, \mathcal{E}^\varepsilon X_{M,\varepsilon}, \mathcal{E}^\varepsilon X_{M,\varepsilon}^\Psi)$ at a chosen time t , for instance $t = 0$. We denote by (φ, X, X^Ψ) a canonical representative of the random variables under consideration and let $\zeta := \varphi - X + X^\Psi$.

Since the energy estimate does not directly gives us a bound on $\zeta_{M,\varepsilon}$ and consequently on $\varphi_{M,\varepsilon}$, we need the following preliminary result.

Theorem 1.3. *Let ρ be a weight such that $\rho^\iota \in L^{4,0}$ for some $\iota \in (0, 1)$. Then*

$$\sup_{M,\varepsilon} \mathbb{E} \|\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0)\|_{H^{1/2-2\kappa,\varepsilon}}^2(\rho^2) \lesssim 1.$$

Proof. Using the first decomposition of $\varphi_{M,\varepsilon}$ together with stationarity of $\varphi_{M,\varepsilon} - X_{M,\varepsilon} = Y_{M,\varepsilon} + \phi_{M,\varepsilon}$ leads to

$$\begin{aligned} \mathbb{E} \|\rho^2(\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0))\|_{H^{1/2-2\kappa,\varepsilon}}^2 &= \frac{1}{T} \int_0^T \mathbb{E} \|\rho^2(\varphi_{M,\varepsilon}(s) - X_{M,\varepsilon}(s))\|_{H^{1/2-2\kappa,\varepsilon}}^2 ds \\ &= \frac{1}{T} \int_0^T \mathbb{E} \|\rho^2(\phi_{M,\varepsilon}(s) + Y_{M,\varepsilon}(s))\|_{H^{1/2-2\kappa,\varepsilon}}^2 ds \\ &\lesssim \frac{1}{T} \int_0^T \mathbb{E} \|\rho^2 \phi_{M,\varepsilon}(s)\|_{H^{1/2-2\kappa,\varepsilon}}^2 ds + \frac{1}{T} \int_0^T \mathbb{E} \|\rho^2 Y_{M,\varepsilon}(s)\|_{H^{1/2-2\kappa,\varepsilon}}^2 ds. \end{aligned} \quad (1.15)$$

The precisely value of T will be chosen below. In view of the energy estimate from Theorem 1.2, we may bound the first term on the right hand side as

$$\frac{1}{T} \int_0^T \mathbb{E} \|\rho^2 \phi_{M,\varepsilon}(s)\|_{H^{1/2-2\kappa,\varepsilon}}^2 ds \leq C_T \mathbb{E} Q(\mathbb{X}_{M,\varepsilon}) + \frac{1}{2T} \mathbb{E} \|\rho^2 \phi_{M,\varepsilon}(0)\|_{L^{2,\varepsilon}}^2.$$

Furthermore, by a Banach fixed point argument we obtain a bound for $Y_{M,\varepsilon}$ over an arbitrary finite time interval $[0, T]$ (see Lemma 4.1 in [4] for more details)

$$\|Y_{M,\varepsilon}\|_{C_T C^{1/2-\kappa,\varepsilon}(\rho^\sigma)} \lesssim \|X_{M,\varepsilon}^\vee\|_{C_T C^{1/2-\kappa,\varepsilon}(\rho^\sigma)}. \quad (1.16)$$

Consequently, we estimate (1.15) as

$$\begin{aligned} & \mathbb{E} \|\rho^2(\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0))\|_{H^{1/2-2\kappa,\varepsilon}}^2 \\ & \lesssim C_T \mathbb{E} Q(\mathbb{X}_{M,\varepsilon}) + \frac{1}{2T} \mathbb{E} \|\rho^2 \phi_{M,\varepsilon}(0)\|_{L^{2,\varepsilon}}^2 + \mathbb{E} \|\rho^\sigma Y_{M,\varepsilon}\|_{C_T C^{1/2-\kappa,\varepsilon}}^2 \\ & \lesssim C_T \mathbb{E} Q(\mathbb{X}_{M,\varepsilon}) + \frac{1}{T} \mathbb{E} \|\rho^2(\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0))\|_{L^{2,\varepsilon}}^2 + \frac{1}{T} \mathbb{E} \|\rho^2 Y_{M,\varepsilon}(0)\|_{L^{2,\varepsilon}}^2 \\ & \lesssim C_T \mathbb{E} Q(\mathbb{X}_{M,\varepsilon}) + \frac{1}{T} \mathbb{E} \|\rho^2(\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0))\|_{L^{2,\varepsilon}}^2 \end{aligned}$$

where we used the fact that $Y_{M,\varepsilon}(0) = -X_{M,\varepsilon}^\vee(0)$ in the last inequality. We see that choosing T large enough, the second term on the right hand side can be absorbed into the left hand side. This computation also determines the time horizon T used in the definition of the polynomial $Q(\mathbb{X}_{M,\varepsilon})$. The proof is complete. $\square$

Theorem 1.4. *The family of joint laws of $(\mathcal{E}^\varepsilon \varphi_{M,\varepsilon}(0), \mathcal{E}^\varepsilon X_{M,\varepsilon}(0), \mathcal{E}^\varepsilon X_{M,\varepsilon}^\vee(0))_{M,\varepsilon}$ is tight on $H^{-1/2-3\kappa}(\rho^{2+\kappa}) \times C^{-1/2-\kappa}(\rho^\sigma) \times C^{1/2-\kappa}(\rho^\sigma)$. Every accumulation point μ satisfies*

$$\mathbb{E}_\mu \|\varphi\|_{H^{-1/2-2\kappa}(\rho^2)}^2 + \mathbb{E}_\mu \|\zeta\|_{H^{1-2\kappa}(\rho^2)}^2 + \mathbb{E}_\mu \|\zeta\|_{B_{4,\infty}^0(\rho^2)}^4 \lesssim 1$$

and the projection of μ onto the first coordinate is the candidate Euclidean Φ_3^4 quantum field theory.

Proof. As a consequence of Theorem 1.3, Lemma A.9 and the uniform bound on $X_{M,\varepsilon}$ we obtain

$$\mathbb{E} \|\mathcal{E}^\varepsilon \varphi_{M,\varepsilon}(0)\|_{H^{-1/2-2\kappa}(\rho^2)}^2 \lesssim 1,$$

which now together with the uniform bound for $X_{M,\varepsilon}^\vee$ yields the desired tightness by compact embeddings of Besov spaces. This uniform estimate carries over to every limit point μ .

It remains to establish the claimed bounds for ζ , for which we make use of its definition (1.14) and the energy estimate from Theorem 1.2. Integrating it in time, applying the decomposition of $\varphi_{M,\varepsilon}$ and using Theorem 1.3 and (1.16) gives for all $t \geq 0$

$$\begin{aligned} & \|\rho^2 \phi_{M,\varepsilon}(t)\|_{L^{2,\varepsilon}}^2 \leq \|\rho^2 \phi_{M,\varepsilon}(0)\|_{L^{2,\varepsilon}}^2 + C_t Q(\mathbb{X}_{M,\varepsilon}) \\ & \lesssim \|\rho^2(\varphi_{M,\varepsilon}(0) - X_{M,\varepsilon}(0))\|_{L^{2,\varepsilon}}^2 + \|\rho^2 Y_{M,\varepsilon}(0)\|_{L^{2,\varepsilon}}^2 + C_t Q(\mathbb{X}_{M,\varepsilon}) \lesssim_t 1. \end{aligned}$$

Moreover,

$$\int_0^T \mathbb{E} \|\phi_{M,\varepsilon}(t)\|_{L^{4,\varepsilon}(\rho)}^4 dt + \int_0^T \mathbb{E} \|\phi_{M,\varepsilon}(t)\|_{H^{1-2\kappa,\varepsilon}(\rho)}^2 dt \lesssim T.$$

In view of the definition of $\zeta_{M,\varepsilon}$ in (1.14), it remains to bound $L_\varepsilon^{-1}[3(U_\varepsilon^\varepsilon[X_{M,\varepsilon}^2]) > Y_{M,\varepsilon}]$. By Schauder estimates, Lemma A.13, paraproduct estimates and (1.16)

$$\begin{aligned} & \mathbb{E} \|L_\varepsilon^{-1}[3(U_\varepsilon^\varepsilon[X_{M,\varepsilon}^2]) > Y_{M,\varepsilon}]\|_{C_T C^{1-\kappa,\varepsilon}(\rho^{2\sigma})}^2 \lesssim \mathbb{E} \|(U_\varepsilon^\varepsilon[X_{M,\varepsilon}^2]) > Y_{M,\varepsilon}\|_{C_T C^{-1-\kappa,\varepsilon}(\rho^{2\sigma})}^2 \\ & \lesssim \mathbb{E} [\|X_{M,\varepsilon}^2\|_{C_T C^{-1-\kappa,\varepsilon}(\rho^\sigma)}^2 \|Y_{M,\varepsilon}\|_{C_T C^{1/2-\kappa,\varepsilon}(\rho^\sigma)}^2] \lesssim 1. \end{aligned}$$

All the above implicit constants are independent of M, ε . Accordingly, the same bounds remain valid under any limit probability measure μ . Since $C^{1-\kappa}(\rho^{2\sigma}) \subset H^{1-2\kappa}(\rho^2)$, the first bound for ζ under μ follows. Similarly, the embedding $L^{4,\varepsilon}(\rho) \subset B_{4,\infty}^0(\rho)$ implies the second bound and the proof is complete. $\square$

Let ν denote the marginal of any limit measure μ corresponding to the variable φ . This is the desired Φ_3^4 measure. It can be shown that it satisfies further physical properties formulated as the so-called Osterwalder–Schrader axioms. For details we refer to Section 5 in [4].

Theorem 1.5. *Any limit measure ν constructed via the procedure in Theorem 1.4 is translation invariant, reflection positive, non-Gaussian and satisfies the distribution property.*

Based on the above construction it is not possible to establish all the OS axioms for Euclidean quantum field theories. The missing axioms are rotation invariance and clustering.

Appendix A

Auxiliary results

The results mentioned here are suitable for tutorials and it is mentioned in the form of exercises in the main text above when each part shall be studied.

A.1 Discrete Besov spaces

We denote by $\Lambda_\varepsilon = (\varepsilon\mathbb{Z})^d$ for $\varepsilon = 2^{-N}$, $N \in \mathbb{N}_0$, the rescaled lattice $\mathbb{Z}^d$ and by $\Lambda_{M,\varepsilon} = \varepsilon\mathbb{Z}^d \cap \mathbb{T}_M^d = \varepsilon\mathbb{Z}^d \cap [-\frac{M}{2}, \frac{M}{2})^d$ its periodic counterpart of size $M > 0$ such that $M/(2\varepsilon) \in \mathbb{N}$. For notational simplicity, we use the convention that the case $\varepsilon = 0$ always refers to the continuous setting. For instance, we denote by Λ_0 the full space $\Lambda_0 = \mathbb{R}^d$ and by $\Lambda_{M,0}$ the continuous torus $\Lambda_{M,0} = \mathbb{T}_M^d$. With the slight abuse of notation, the parameter ε is always taken either of the form $\varepsilon = 2^{-N}$ for some $N \in \mathbb{N}_0$, $N \geq N_0$, for certain $N_0 \in \mathbb{N}_0$ that will be chosen as a consequence of Lemma A.10 below, or $\varepsilon = 0$. Various proofs below will be formulated generally for $\varepsilon \in \mathcal{A} := \{0, 2^{-N}; N \in \mathbb{N}_0, N \geq N_0\}$ and it is understood that the case $\varepsilon = 0$ or alternatively $N = \infty$ refers to the continuous setting. All the proportionality constants, unless explicitly stated, are independent of M, ε .

Denote $\hat{\Lambda}_\varepsilon := (\varepsilon^{-1}\mathbb{T})^d$. For $f \in \ell^1(\Lambda_\varepsilon)$ and $g \in L^1(\hat{\Lambda}_\varepsilon)$ we define the Fourier and the inverse Fourier transform as

$$\mathcal{F}f(k) := \varepsilon^d \sum_{x \in \Lambda_\varepsilon} f(x) e^{-2\pi i k \cdot x}, \quad \mathcal{F}^{-1}g(x) := \int_{\hat{\Lambda}_\varepsilon} g(k) e^{2\pi i k \cdot x} dk,$$

where $k \in \hat{\Lambda}_\varepsilon$ and $x \in \Lambda_\varepsilon$. These definitions can be extended to discrete Schwartz distributions in a natural way, we refer to [6] for more details. In general, we do not specify on which lattice the Fourier transform is taken as it will be clear from the context.

Consider a smooth dyadic partition of unity $(\varphi_j)_{j \geq -1}$ such that φ_{-1} is supported in a ball around 0 of radius $\frac{1}{2}$, φ_0 is supported in an annulus, $\varphi_j(\cdot) = \varphi_0(2^{-j}\cdot)$ for $j \geq 0$ and if $|i - j| > 1$ then $\text{supp } \varphi_i \cap \text{supp } \varphi_j = \emptyset$. For the definition of Besov spaces on the lattice Λ_ε for $\varepsilon = 2^{-N}$, we introduce a suitable periodic partition of unity on $\hat{\Lambda}_\varepsilon$ as follows

$$\varphi_j^\varepsilon(k) := \begin{cases} \varphi_j(k), & j < N - \mathcal{J}, \\ 1 - \sum_{j < N - \mathcal{J}} \varphi_j(k), & j = N - \mathcal{J}, \end{cases} \quad (\text{A.1})$$

where $k \in \hat{\Lambda}_\varepsilon$ and the parameter $\mathcal{J} \in \mathbb{N}_0$, whose precise value will be chosen below independently on $\varepsilon \in \mathcal{A}$, satisfies $0 \leq N - \mathcal{J} \leq \mathcal{J}_\varepsilon := \inf \{j : \text{supp } \varphi_j \not\subseteq [-\varepsilon^{-1}/2, \varepsilon^{-1}/2)^d\} \rightarrow \infty$ as $\varepsilon \rightarrow 0$. We note that by construction there exists $\ell \in \mathbb{Z}$ independent of $\varepsilon = 2^{-N}$ such that $\mathcal{J}_\varepsilon = N - \ell$.

Then (A.1) yields a periodic partition of unity on $\hat{\Lambda}_\varepsilon$. The reason for choosing the upper index as $N - \mathcal{J}$ and not the maximal choice $\mathcal{J}_\varepsilon$ will become clear in Lemma A.10 below, where it allows us to define suitable localization operators needed for our analysis. The choices of parameters N_0 and $\mathcal{J}$ are related in the following way: A given partition of unity $(\varphi_j)_{j \geq -1}$ determines the parameters $\mathcal{J}_\varepsilon$ in the form $\mathcal{J}_\varepsilon = N - \ell$ for some $\ell \in \mathbb{Z}$. By the condition $N - \mathcal{J} \leq \mathcal{J}_\varepsilon$ we obtain the first lower bound on $\mathcal{J}$. Then Lemma A.10 yields a (possibly larger) value of $\mathcal{J}$ which is fixed throughout the paper. Finally, the condition $0 \leq N - \mathcal{J}$ implies the necessary lower bound N_0 for N , or alternatively the upper bound for $\varepsilon = 2^{-N} \leq 2^{-N_0}$ and defines the set $\mathcal{A}$. We stress that once the parameters $\mathcal{J}, N_0$ are chosen, they remain fixed.

Then (A.1) yields a periodic partition of unity on $\hat{\Lambda}_\varepsilon$. The reason for choosing the upper index as $N - \mathcal{J}$ and not the maximal choice $\mathcal{J}_\varepsilon$ will become clear in Lemma A.10 below, where it allows us to define suitable localization operators needed for our analysis. The choices of parameters N_0 and $\mathcal{J}$ are related in the following way: A given partition of unity $(\varphi_j)_{j \geq -1}$ determines the parameters $\mathcal{J}_\varepsilon$ in the form $\mathcal{J}_\varepsilon = N - \ell$ for some $\ell \in \mathbb{Z}$. By the condition $N - \mathcal{J} \leq \mathcal{J}_\varepsilon$ we obtain the first lower bound on $\mathcal{J}$. Then Lemma A.10 yields a (possibly larger) value of $\mathcal{J}$ which is fixed throughout the paper. Finally, the condition $0 \leq N - \mathcal{J}$ implies the necessary lower bound N_0 for N , or alternatively the upper bound for $\varepsilon = 2^{-N} \leq 2^{-N_0}$ and defines the set $\mathcal{A}$. We stress that once the parameters $\mathcal{J}, N_0$ are chosen, they remain fixed.

Remark that according to our convention, $(\varphi_j^0)_{j \geq -1}$ denotes the original partition of unity $(\varphi_j)_{j \geq -1}$ on $\mathbb{R}^d$, which can be also read from (A.1) using the fact that for $\varepsilon = 0$ we have $\mathcal{J}_\varepsilon = \infty$.

Now we may define the Littlewood–Paley blocks for distributions on Λ_ε by

$$\Delta_j^\varepsilon f := \mathcal{F}^{-1}(\varphi_j^\varepsilon \mathcal{F}f),$$

which leads us to the definition of weighted Besov spaces. In the sequel, ρ denotes a polynomial weight of the form

$$\rho(x) = \langle h x \rangle^{-\varsigma} = (1 + |h x|^2)^{-\varsigma/2} \quad (\text{A.2})$$

for some $\varsigma > 0$ and $h > 0$. The constant h shall be fixed in the weighted energy estimate in Section 1.2.2 in order to produce a small bound for certain terms. Such weights satisfy the admissibility condition $\rho(x)/\rho(y) \lesssim \rho^{-1}(x - y)$ for all $x, y \in \mathbb{R}^d$. For $\alpha \in \mathbb{R}$, $p, q \in [1, \infty]$ and $\varepsilon \in [0, 1]$ we define the weighted Besov spaces on Λ_ε by the norm

$$\|f\|_{B_{p,q}^{\alpha,\varepsilon}(\rho)} = \left(\sum_{-1 \leq j \leq N - \mathcal{J}} 2^{\alpha j q} \|\Delta_j^\varepsilon f\|_{L^{p,\varepsilon}(\rho)}^q \right)^{1/q} = \left(\sum_{-1 \leq j \leq N - \mathcal{J}} 2^{\alpha j q} \|\rho \Delta_j^\varepsilon f\|_{L^{p,\varepsilon}}^q \right)^{1/q},$$

where $L^{p,\varepsilon}$ for $\varepsilon \in \mathcal{A} \setminus \{0\}$ stands for the L^p space on Λ_ε given by the norm

$$\|f\|_{L^{p,\varepsilon}} = \left(\varepsilon^d \sum_{x \in \Lambda_\varepsilon} |f(x)|^p \right)^{1/p}$$

(with the usual modification if $p = \infty$). Analogously, we may define the weighted Besov spaces for explosive polynomial weights of the form ρ^{-1} . Note that if $\varepsilon = 0$ then $B_{p,q}^{\alpha,\varepsilon}(\rho)$ is the classical weighted Besov space $B_{p,q}^\alpha(\rho)$. In the sequel, we also employ the following notations

$$C^{\alpha,\varepsilon}(\rho) := B_{\infty,\infty}^{\alpha,\varepsilon}(\rho), \quad H^{\alpha,\varepsilon}(\rho) := B_{2,2}^{\alpha,\varepsilon}(\rho).$$

We will frequently use the following auxiliary results whose proofs can be found in Appendix A.1 in [4].

Lemma A.1. *Let $\alpha \in \mathbb{R}$, $p, q \in [1, \infty]$. Fix $n > |\alpha|$ and assume that ρ is a weight such that*

$$\|\rho\|_{B_{\infty,\infty}^{n+1,\varepsilon}(\rho^{-1})} + \|\rho^{-1}\|_{B_{\infty,\infty}^{n+1,\varepsilon}(\rho)} \lesssim 1$$

uniformly in ε . Then

$$\|f\|_{B_{p,q}^{\alpha,\varepsilon}(\rho)} \sim \|\rho f\|_{B_{p,q}^{\alpha,\varepsilon}},$$

where the proportionality constant does not depend on ε .

Lemma A.2. *Let $\alpha \in \mathbb{R}$, $p, p', q, q' \in [1, \infty]$ such that p, p' and q, q' are conjugate exponents. Let ρ be a weight as in Lemma A.1. Then*

$$\langle f, g \rangle_\varepsilon \lesssim \|f\|_{B_{p,q}^{\alpha,\varepsilon}(\rho)} \|g\|_{B_{p',q'}^{-\alpha,\varepsilon}(\rho^{-1})}$$

with a proportionality constant independent of ε . Consequently, $B_{p',q'}^{-\alpha,\varepsilon}(\rho^{-1}) \subset (B_{p,q}^{\alpha,\varepsilon}(\rho^{-1}))^*$.

Lemma A.3. Let $\varepsilon \in \mathcal{A}$. Let $\alpha, \alpha_0, \alpha_1, \beta, \beta_0, \beta_1 \in \mathbb{R}$, $p, p_0, p_1, q, q_0, q_1 \in [1, \infty]$ and $\theta \in [0, 1]$ such that

$$\alpha = \theta \alpha_0 + (1 - \theta) \alpha_1, \quad \beta = \theta \beta_0 + (1 - \theta) \beta_1, \quad \frac{1}{p} = \frac{\theta}{p_0} + \frac{1 - \theta}{p_1}, \quad \frac{1}{q} = \frac{\theta}{q_0} + \frac{1 - \theta}{q_1}.$$

Then

$$\|f\|_{B_{p,q}^{\alpha,\varepsilon}(\rho^\beta)} \leq \|f\|_{B_{p_0,q_0}^{\alpha_0,\varepsilon}(\rho^{\beta_0})}^\theta \|f\|_{B_{p_1,q_1}^{\alpha_1,\varepsilon}(\rho^{\beta_1})}^{1-\theta}.$$

Lemma A.4. Let $\varepsilon \in \mathcal{A}$ and let $\beta > 0$. Then

$$L^{2,\varepsilon}(\rho) = B_{2,2}^{0,\varepsilon}(\rho), \quad L^{4,\varepsilon}(\rho) \subset B_{4,\infty}^{0,\varepsilon}(\rho)$$

and the proportional constants do not depend on ε .

Lemma A.5. Let $\kappa \in (0, 1)$, $p \in [1, \infty]$ and let ρ be a polynomial weight

$$\|f\|_{B_{p,p}^{1-\kappa,\varepsilon}(\rho)} \lesssim \|f\|_{B_{p,p}^{-\kappa,\varepsilon}(\rho)} + \|\nabla_\varepsilon f\|_{B_{p,p}^{-\kappa,\varepsilon}(\rho)},$$

where the proportionality constant does not depend on ε .

Lemma A.6. Let $\varepsilon \in \mathcal{A}$ and let $\iota > 0$. Let ρ be a weight such that $\rho' \in L^{4,0}$. Then

$$\|\rho^{1+\iota} f\|_{L^{2,\varepsilon}} \lesssim \|\rho f\|_{L^{4,\varepsilon}},$$

where the proportionality constant does not depend on ε .

Lemma A.7. Let $\alpha > 0$. Let ρ_1, ρ_2 be weights. Then for every $\beta > 0$

$$\|f^2\|_{B_{1,1}^{\alpha,\varepsilon}(\rho_1 \rho_2)} \lesssim \|f\|_{L^{2,\varepsilon}(\rho_1)} \|f\|_{H^{\alpha+2\beta,\varepsilon}(\rho_2)},$$

$$\|f^3\|_{B_{1,1}^{\alpha,\varepsilon}(\rho_1^2 \rho_2)} \lesssim \|f\|_{L^{4,\varepsilon}(\rho_1)}^2 \|f\|_{H^{\alpha+2\beta,\varepsilon}(\rho_2)},$$

where the proportionality constants do not depend on ε .

The following Besov embedding can be found in Lemma 2.22 in [6].

Lemma A.8. Let ρ_1, ρ_2 be weights satisfying $\rho_2 \lesssim \rho_1$. Let $\alpha \in \mathbb{R}$, $1 \leq p_1 \leq p_2 \leq \infty$, $1 \leq q_1 \leq q_2 \leq \infty$. Then the following embedding is continuous

$$B_{p_1,q_1}^{\alpha,\varepsilon}(\rho_1) \subset B_{p_2,q_2}^{\alpha-d(1/p_1-1/p_2),\varepsilon}(\rho_2)$$

and the proportionality constants do not depend on ε .

A.2 Extension operators

In order to construct the Euclidean quantum field theory as a limit of lattice approximations, we need a suitable extension operator that allows to extend distributions defined on the lattice Λ_ε to the full space $\mathbb{R}^d$. To this end, we fix a smooth, compactly supported and radially symmetric nonnegative function $w \in C_c^\infty(\mathbb{R}^d)$ such that $\text{supp } w \subset B_{1/2}$ where $B_{1/2} \subset \mathbb{R}^d$ is the ball centered at 0 with radius $1/2$ and $\int_{\mathbb{R}^d} w(x) \, dx = 1$. Let $w^\varepsilon(\cdot) := \varepsilon^{-d} w(\varepsilon^{-1} \cdot)$ and define the extension operator $\mathcal{E}^\varepsilon$ by

$$\mathcal{E}^\varepsilon f := w^\varepsilon \star_\varepsilon f, \quad f \in \mathcal{S}'(\Lambda_\varepsilon).$$

With a slight abuse of notation we used the same notation $*_\varepsilon$ as for the convolution on the lattice Λ_ε to denote the operation

$$(w^\varepsilon *_\varepsilon f)(x) := \varepsilon^d \sum_{y \in \Lambda_\varepsilon} w^\varepsilon(x-y) f(y), \quad x \in \mathbb{R}^d,$$

which defines a function on the full space $\mathbb{R}^d$. The following result is proved in Section A.4 in [4].

Lemma A.9. *Let $\alpha \in \mathbb{R}$, $p, q \in [1, \infty]$ and let ρ be a weight. Then the operators*

$$\mathcal{E}^\varepsilon: B_{p,q}^{\alpha,\varepsilon}(\rho) \rightarrow B_{p,q}^\alpha(\rho)$$

are bounded uniformly in ε .

A.3 Paraproducts

We employ the tools from paracontrolled calculus as introduced in [2], the reader is also referred to [1] for further details. We shall freely use the decomposition $fg = f < g + f \circ g + f > g$, where $f > g = g > f$ and $f \circ g$, respectively, stands for the paraproduct of f and g and the corresponding resonant term, defined in terms of Littlewood–Paley decomposition. More precisely, for $f, g \in \mathcal{S}'(\Lambda_\varepsilon)$ we let

$$f < g := \sum_{1 \leq i, j \leq N-j, i < j-1} \Delta_i^\varepsilon f \Delta_j^\varepsilon g, \quad f \circ g := \sum_{1 \leq i, j \leq N-j, i \sim j} \Delta_i^\varepsilon f \Delta_j^\varepsilon g.$$

We also employ the notations $f \leq g := f < g + f \circ g$. For notational simplicity, we do not stress the dependence of the paraproduct and the resonant term on ε in the sequel. These paraproducts satisfy the usual estimates uniformly in ε , see e.g. [6], Lemma 4.2, which can be naturally extended to general $B_{p,q}^{\alpha,\varepsilon}(\rho)$ Besov spaces as in [7], Theorem 3.17.

A.4 Localizers

As the next step, we introduce another equivalent formulation of the weighted Besov spaces $B_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ in terms of suitable point evaluation of the Littlewood–Paley decomposition. First, for $j \in \mathbb{N}_0$ such that $N-j \leq j_\varepsilon$, $\alpha \in \mathbb{R}$ and $\varepsilon \in \mathcal{A}$ we define the Besov space $b_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ of sequences $\lambda = (\lambda_{j,m})_{-1 \leq j \leq N-j, m \in \mathbb{Z}^d}$ by the norm

$$\|\lambda\|_{b_{\infty,\infty}^{\alpha,\varepsilon}(\rho)} := \sup_{-1 \leq j \leq N-j} 2^{\alpha j} \sup_{m \in \mathbb{Z}^d} \rho(2^{-j-j} m) |\lambda_{j,m}|.$$

Note that we do not stress the dependence of $b_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ on the parameter j as in the sequel we only consider one fixed j for all $\varepsilon \in \mathcal{A}$ given by Lemma A.10 below. The next result shows the desired equivalence and can be found in Appendix A.2 in [4].

Lemma A.10. *Let $\alpha \in \mathbb{R}$, $\varepsilon \in \mathcal{A}$ and let ρ be a weight. There exists $j \in \mathbb{N}_0$ (independent of ε) with the following property: $f \in B_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ if and only if it is represented by $\lambda = (\lambda_{j,m})_{-1 \leq j \leq N-j, m \in \mathbb{Z}^d} \in b_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ such that*

$$\|f\|_{B_{\infty,\infty}^{\alpha,\varepsilon}(\rho)} \sim \|\lambda\|_{b_{\infty,\infty}^{\alpha,\varepsilon}(\rho)}, \tag{A.3}$$

where the proportionality constants do not depend on ε . In particular, given $f \in B_{\infty,\infty}^{\alpha,\varepsilon}(\rho)$ the coefficients λ are defined by

$$\lambda_{j,m}(f) := \Delta_j^\varepsilon f(2^{-j-j} m), \quad -1 \leq j \leq N-j, \quad m \in \mathbb{Z}^d, \tag{A.4}$$

and given $\lambda \in b_{\infty, \infty}^{\alpha, \varepsilon}(\rho)$ the distribution f is recovered via the formula

$$f = \sum_{-1 \leq j \leq N-j} \mathcal{F}^{-1}(\mathcal{F}_{2^{-j-j}\mathbb{Z}^d}(\lambda_{j, \cdot})), \quad (\text{A.5})$$

where $\mathcal{F}_{2^{-j-j}\mathbb{Z}^d}$ denotes the Fourier transform on the lattice $2^{-j-j}\mathbb{Z}^d$.

Remark A.11. Throughout the paper, the parameter $j \in \mathbb{N}_0$ is fixed as in Lemma A.10. Consequently, from the condition $0 \leq N - j$ we obtain the necessary lower bound N_0 for N , or alternatively the upper bound for $\varepsilon = 2^{-N} \leq 2^{-N_0}$ and defines the set $\mathcal{A}$. These parameters remain fixed throughout the text.

Remark A.12. Note that the formulas (A.4), (A.5) depend on the chosen partition of unity $(\varphi_j)_{j \geq -1}$ and our construction of the associated periodic partitions of unity on $\hat{\Lambda}_\varepsilon$ via (A.1).

It follows from the previous lemma that we may identify $f \in B_{\infty, \infty}^{\alpha, \varepsilon}(\rho)$ with its coefficients $(\lambda_{j, m}(f))_{-1 \leq j \leq N-j, m \in \mathbb{Z}^d} \in b_{\infty, \infty}^{\alpha, \varepsilon}(\rho)$. This consideration leads us to the definition of localization operators needed for the analysis of the Φ_3^4 model. Although the principle idea is similar to Section 2.3 in [3], we present a different definition of the localizers here. It is based on the equivalent description of the Besov spaces from Lemma A.10 and is better suited for the discrete setting.

Given $(L_k)_{k \geq -1} \subset (0, \infty)$ and $f \in \mathcal{S}'(\Lambda_\varepsilon)$ we define

$$U_{>}^\varepsilon f := (\lambda_{j, m}(U_{>}^\varepsilon f))_{-1 \leq j \leq N-j, m \in \mathbb{Z}^d}, \quad U_{\leq}^\varepsilon f := (\lambda_{j, m}(U_{\leq}^\varepsilon f))_{-1 \leq j \leq N-j, m \in \mathbb{Z}^d}$$

where

$$\lambda_{j, m}(U_{>}^\varepsilon f) := \begin{cases} \lambda_{j, m}(f), & \text{if } |m| \sim 2^k \text{ and } j > L_k \text{ for some } k \in \{-1, 0, 1, \dots\}, \\ 0, & \text{otherwise,} \end{cases}$$

$$\lambda_{j, m}(U_{\leq}^\varepsilon f) := \begin{cases} \lambda_{j, m}(f), & \text{if } |m| \sim 2^k \text{ and } j \leq L_k \text{ for some } k \in \{-1, 0, 1, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

We observe that by definition $f = U_{>}^\varepsilon f + U_{\leq}^\varepsilon f$ and the localizers $U_{>}^\varepsilon, U_{\leq}^\varepsilon$ will only depend on ε through the cut-off of the coefficients λ (and consequently on the construction of the partition of unity on $\hat{\Lambda}_\varepsilon$, cf. Remark A.12), whereas the sequence $(L_k)_{k \geq -1}$ will be chosen uniformly for all $\varepsilon \in \mathcal{A}$. The following result can be found in Appendix A.2 in [4].

Lemma A.13. Let ρ be a weight. Let $\alpha, \beta, \gamma \in \mathbb{R}$ and $a, b, c \in \mathbb{R}$ such that $\alpha < \beta < \gamma$, $a < b < c$ and $r := (b - a) / (\beta - \alpha) = (c - b) / (\gamma - \beta) > 0$. Let $L > 0$ be given. There exists a sequence $(L_k)_{k \geq -1}$ defining the above localizers such that

$$|U_{>}^\varepsilon f|_{B_{\infty, \infty}^{\alpha, \varepsilon}(\rho^a)} \lesssim 2^{-(\beta - \alpha)L} \|f\|_{B_{\infty, \infty}^{\beta, \varepsilon}(\rho^b)},$$

$$|U_{\leq}^\varepsilon f|_{B_{\infty, \infty}^{\gamma, \varepsilon}(\rho^c)} \lesssim 2^{(\gamma - \beta)L} \|f\|_{B_{\infty, \infty}^{\beta, \varepsilon}(\rho^b)},$$

where the proportionality constants do not depend on $\varepsilon \in \mathcal{A}$. Moreover, the sequence $(L_k)_{k \geq -1}$ depends only on L, ρ and the ratio r .

A.5 Approximate duality and commutators

In this section we define various commutators and establish suitable bounds. The proofs can be found in Appendix A.3 in [4].

First, we denote by C_ε the operator introduced in Lemma 4.3 [6], which for smooth functions satisfies

$$C_\varepsilon(f, g, h) = h \circ (f \prec g) - f(h \circ g). \quad (\text{A.6})$$

We recall that if $p, p_1, p_2 \in [1, \infty]$ and $\alpha, \beta, \gamma \in \mathbb{R}$ are such that $\frac{1}{p} = \frac{1}{p_1} + \frac{1}{p_2}$, $\alpha + \beta + \gamma > 0$ and $\beta + \gamma \neq 0$, then the following bound holds

$$\|C_\varepsilon(f, g, h)\|_{B_{p, \infty}^{\beta+\gamma, \varepsilon}(\rho_1 \rho_2 \rho_3)} \lesssim \|f\|_{B_{p_1, \infty}^{\alpha, \varepsilon}(\rho_1)} \|g\|_{B_{\infty, \infty}^{\beta, \varepsilon}(\rho_2)} \|h\|_{B_{p_2, \infty}^{\gamma, \varepsilon}(\rho_3)}. \quad (\text{A.7})$$

The following result says that $g \succ$ is an approximate adjoint of $g \circ$ in a suitable sense.

Lemma A.14. *Let $\varepsilon \in \mathcal{A}$. Let $\alpha, \beta, \gamma \in \mathbb{R}$ be such that $\alpha, \gamma > 0$, $\beta + \gamma < 0$ and $\alpha + \beta + \gamma > 0$ and let ρ_1, ρ_2, ρ_3 be weights and let $\rho = \rho_1 \rho_2 \rho_3$. There exists a bounded trilinear operator*

$$D_{\rho, \varepsilon}(f, g, h): H^{\alpha, \varepsilon}(\rho_1) \times C^{\beta, \varepsilon}(\rho_2) \times H^{\gamma, \varepsilon}(\rho_3) \rightarrow \mathbb{R}$$

such that

$$|D_{\rho, \varepsilon}(f, g, h)| \lesssim \|f\|_{H^{\alpha, \varepsilon}(\rho_1)} \|g\|_{C^{\beta, \varepsilon}(\rho_2)} \|h\|_{H^{\gamma, \varepsilon}(\rho_3)}$$

where the proportionality constant is independent of ε , and for smooth functions we have

$$D_{\rho, \varepsilon}(f, g, h) = \langle \rho f, g \circ h \rangle_\varepsilon - \langle \rho(f \prec g), h \rangle_\varepsilon.$$

Next, we present several commutator estimates. To this end, Δ_ε denotes the discrete Laplacian on Λ_ε and we define the corresponding elliptic and parabolic operators by $Q_\varepsilon := m^2 - \Delta_\varepsilon$ and $\mathcal{L}_\varepsilon := \partial_t + Q_\varepsilon$, where $m^2 > 0$.

Lemma A.15. *Let $\varepsilon \in \mathcal{A}$. Let $\alpha, \beta, \gamma \in \mathbb{R}$ such that $\alpha \in (0, 1)$, $\beta + \gamma + 2 < 0$ and $\alpha + \beta + \gamma + 2 > 0$. Let ρ_1, ρ_2, ρ_3 be space weights and let ρ_4, ρ_5, ρ_6 be space-time weights. Then there exist bounded trilinear operators*

$$\begin{aligned} \tilde{C}_\varepsilon: H^{\alpha, \varepsilon}(\rho_1) \times C^{\beta, \varepsilon}(\rho_2) \times C^{\gamma+\delta, \varepsilon}(\rho_3) &\rightarrow H^{\beta+\gamma+2, \varepsilon}(\rho_1 \rho_2 \rho_3), \\ \bar{C}_\varepsilon: C_T C^{\alpha, \varepsilon}(\rho_4) \times C_T C^{\beta, \varepsilon}(\rho_5) \times C_T C^{\gamma+\delta, \varepsilon}(\rho_6) &\rightarrow C_T C^{\beta+\gamma+2, \varepsilon}(\rho_4 \rho_5 \rho_6) \end{aligned}$$

such that for every $\delta > 0$

$$\begin{aligned} \|\tilde{C}_\varepsilon(f, g, h)\|_{H^{\beta+\gamma+2, \varepsilon}(\rho_1 \rho_2 \rho_3)} &\lesssim \|f\|_{H^{\alpha, \varepsilon}(\rho_1)} \|g\|_{C^{\beta, \varepsilon}(\rho_2)} \|h\|_{C^{\gamma+\delta, \varepsilon}(\rho_3)}, \\ \|\bar{C}_\varepsilon(f, g, h)\|_{C_T C^{\beta+\gamma+2, \varepsilon}(\rho_4 \rho_5 \rho_6)} &\lesssim (\|f\|_{C_T C^{\alpha, \varepsilon}(\rho_4)} + \|f\|_{C_T^{1/2} L^{\infty, \varepsilon}(\rho_4)}) \|g\|_{C_T C^{\beta, \varepsilon}(\rho_5)} \|h\|_{C_T C^{\gamma+\delta, \varepsilon}(\rho_6)}, \end{aligned}$$

where the proportionality constants are independent of ε , and for smooth functions we have

$$\begin{aligned} \tilde{C}_\varepsilon(f, g, h) &= h \circ Q_\varepsilon^{-1}(f \prec g) - f(h \circ Q_\varepsilon^{-1}g), \\ \bar{C}_\varepsilon(f, g, h) &= h \circ \mathcal{L}_\varepsilon^{-1}(f \prec g) - f(h \circ \mathcal{L}_\varepsilon^{-1}g). \end{aligned} \quad (\text{A.8})$$

Bibliography

- [1] H. Bahouri, J.-Y. Chemin, and R. Danchin. *Fourier Analysis and Nonlinear Partial Differential Equations*. Springer, jan 2011.
- [2] M. Gubinelli, P. Imkeller, and N. Perkowski. Paracontrolled distributions and singular PDEs. *Forum of Mathematics. Pi*, 3:0, 2015.
- [3] Massimiliano Gubinelli and Martina Hofmanová. Global solutions to elliptic and parabolic Φ^4 models in Euclidean space. *Comm. Math. Phys.*, 368(3):1201–1266, 2019.
- [4] Massimiliano Gubinelli and Martina Hofmanová. A PDE construction of the Euclidean ϕ_s^4 quantum field theory. *Comm. Math. Phys.*, 384(1):1–75, 2021.
- [5] R. Khasminskii. *Stochastic stability of differential equations*, volume 66. Springer Science & Business Media, 2011.
- [6] Jörg Martin and Nicolas Perkowski. Paracontrolled distributions on Bravais lattices and weak universality of the 2d parabolic Anderson model. *Ann. Inst. Henri Poincaré Probab. Stat.*, 55(4):2058–2110, 2019.
- [7] J.-C. Mourrat and H. Weber. Global well-posedness of the dynamic Φ^4 model in the plane. *The Annals of Probability*, 45(4):2398–2476, jul 2017.